.things that Matter

Risk Assessment & Mitigation Plan

Document Information	
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Company Name	**.things that Matter**
Document Type	**Risk Assessment**
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1. Risk Management Overview

1.1 Risk Assessment Framework

Probability	Definition
High (H)	**>70% likely to occur**
Medium (M)	**30-70% likely to occur**
Low (L)	**<30% likely to occur**

Impact	Definition
Critical	**Project failure, major pivot required**
High	**Significant delay (>4 weeks), budget overrun >20%**
Medium	**Moderate delay (2-4 weeks), budget impact 10-20%**
Low	**Minor delay (<2 weeks), minimal budget impact**

1.2 Risk Matrix

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              | Low Impact | Medium Impact | High Impact | Critical |
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High Prob     |   Monitor  |    Mitigate   |   Mitigate  |  Avoid   |
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Medium Prob   |   Accept   |    Monitor    |   Mitigate  | Mitigate |
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Low Prob      |   Accept   |    Accept     |   Monitor   | Mitigate |
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2. Risk Register

2.1 Technical Risks

ID	Risk	Probability	Impact	Score	Category
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T01	**Steganography fails on major platforms**	**Medium**	**Critical**	**High**	**Technical**
T02	**CLIP API becomes too expensive**	**Low**	**High**	**Medium**	**Technical**
T03	**FlatBuffers schema design flaws**	**Low**	**High**	**Medium**	**Technical**
T04	**Ed25519 implementation vulnerability**	**Low**	**Critical**	**Medium**	**Security**
T05	**Browser extension rejected by Chrome**	**Low**	**High**	**Medium**	**Technical**
T06	**WASM performance insufficient**	**Low**	**Medium**	**Low**	**Technical**

T07	**Rust learning curve steeper than expected**	**Medium**	**Low**	**Low**	**Technical**

T08	**Cloudflare R2 service disruption**	**Low**	**Medium**	**Low**	**Infrastructure**

2.2 Business Risks

ID	Risk	Probability	Impact	Score	Category
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B01	**No merchant adoption**	**Medium**	**Critical**	**High**	**Business**
B02	**Budget overrun**	**Medium**	**High**	**High**	**Financial**
B03	**Solo developer burnout**	**Medium**	**High**	**High**	**Resource**
B04	**Competitor launches similar product**	**Low**	**Medium**	**Low**	**Market**
B05	**Legal challenge to steganography**	**Low**	**High**	**Medium**	**Legal**
B06	**Scope creep**	**High**	**Medium**	**High**	**Management**

2.3 Operational Risks

ID	Risk	Probability	Impact	Score	Category
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O01	**Cursor Pro limits slow development**	**Medium**	**Medium**	**Medium**	**Operational**
O02	**Free tier services become insufficient**	**Medium**	**Medium**	**Medium**	**Infrastructure**
O03	**Key dependency (Replicate) changes API**	**Low**	**High**	**Medium**	**Dependency**
O04	**Data loss (code, artifacts)**	**Low**	**Critical**	**Medium**	**Operational**
O05	**Security breach before audit**	**Low**	**Critical**	**Medium**	**Security**

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3. Detailed Risk Analysis

3.1 T01: Steganography Platform Failure

Risk: Steganographic payloads do not survive on major social platforms.

Attribute	**Value**
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Probability	**Medium (40-60%)**
Impact	**Critical**
Risk Score	**HIGH**

Risk Owner	CTO

Scenario: - Instagram, Twitter, Facebook all strip XMP metadata and recompress images - EOF appended data destroyed during platform processing - Survival rate <50% across tested platforms

Indicators: - Phase 0 testing shows <50% survival - Major platforms (Instagram, Twitter) show 0% survival - Only minor platforms (Discord, Telegram) preserve data

Mitigation Strategy:

Strategy	Action	Effort
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Validate Early	**Phase 0 testing on 20 platforms before committing**	**4 weeks**
Hybrid Distribution	**If steg fails, prioritize matter:// URIs**	**Parallel dev**
Metadata Focus	**If EOF fails, rely on XMP gist only**	**Design change**
Platform-Specific	**Different strategies per platform**	**+2 weeks**

Contingency Plan: If steganography survival <50%: 1. Pivot to matter:// URI as primary distribution 2. Use polyglot images only for platforms that support them 3. Build URL shortener service for matter:// links 4. Focus on "link in bio" use cases

Residual Risk: Medium (after mitigation)

3.2 B01: No Merchant Adoption

Risk: Merchants do not see value in .thing files and refuse to onboard.

Attribute	Value

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Probability	**Medium (30-50%)**

Impact	**Critical**

Risk Score	**HIGH**

Risk Owner	**CEO**

Scenario: - Merchants don't understand the value proposition - Integration effort seems too high - No clear ROI demonstrated - Competition from established platforms

Indicators: - <3 merchants interested after 2 months outreach - Beta merchants churn before launch - Feedback indicates confusion about value

Mitigation Strategy:

Strategy	**Action**	**Effort**
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Developer First	**Target developers who build merchant tools**	**Strategy shift**
Open Source	**Release compiler open-source for community adoption**	**Month 3**
Demo-Driven	**Build compelling demos before outreach**	**Ongoing**
Niche Focus	**Target specific verticals (NFTs, digital goods)**	**Research**

Contingency Plan: If merchant adoption fails: 1. Pivot to B2D (business-to-developer) model 2. Open-source everything immediately 3. Focus on building community 4. Seek partnership with e-commerce platforms (Shopify, WooCommerce)

Residual Risk: Medium (after mitigation)

3.3 B02: Budget Overrun

Risk: Actual costs exceed R72,000 budget.

Attribute	Value
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Probability	**Medium (40-50%)**
Impact	**High**
Risk Score	**HIGH**
Risk Owner	**CTO + CEO**

Scenario: - CLIP API costs higher than projected - Need to upgrade from free tiers earlier - Unexpected tools or services required - 5% buffer insufficient

Indicators: - Monthly spend exceeds projection by >20% - CLIP usage higher than estimated - Free tier limits hit early - Contingency depleted by Month 6

Mitigation Strategy:

Strategy	Action	Effort
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Track Weekly	**Monitor expenses vs. budget weekly**	**Ongoing**

Cost Alerts	**Set up billing alerts on all services**	**1 day**

Usage Optimization	**Cache CLIP results, batch requests**	**Design**

Free Tier Discipline	**Stay on free tiers as long as possible**	**Discipline**

Budget Safeguards:

Expense	Budget	Hard Limit	Action if Exceeded
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CLIP API	**R21,600**	**R25,000**	**Reduce batch sizes**
R2 Storage	**R7,560**	**R10,000**	**Compress more aggressively**
Development Tools	**R6,048**	**R7,000**	**Drop optional tools**
Total	**R72,000**	**R80,000**	**Pause non-critical work**

Contingency Plan: If budget exhausted: 1. Pause development, seek additional funding 2. Reduce scope to MVP only 3. Release open-source, seek community contributions 4. CEO to explore bridge funding

Residual Risk: Medium (after mitigation)

3.4 B03: Solo Developer Burnout

Risk: Single developer experiences burnout, affecting delivery.

Attribute	Value

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Probability	**Medium (40-50%)**

Impact	**High**

Risk Score	**HIGH**

Risk Owner	**CTO + CEO**

Scenario: - 11-month solo development becomes overwhelming - No time off, constant pressure - Health issues arise - Quality degrades due to fatigue

Indicators: - Missing weekly commitments - Declining code quality - Communication becomes sparse - Expressed frustration or exhaustion

Mitigation Strategy:

Strategy	**Action**	**Effort**
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Sustainable Pace	**Strict 30 hrs/week limit**	**Discipline**
Weekly Rest	**No coding on Sundays**	**Policy**
Milestone Celebrations	**Acknowledge each 15% increment**	**Morale**
Open Communication	**CEO checks in on wellbeing**	**Weekly**

Burnout Prevention Checklist:

- [ ] 30 hours/week maximum - [ ] One full day off per week - [ ] No late-night coding sessions - [ ] Exercise/breaks scheduled - [ ] Vacation days planned (at least 1 week) - [ ] CEO check-ins on

wellbeing

Contingency Plan: If burnout occurs: 1. Immediate 1-2 week break 2. Adjust timeline (add 1-2 months) 3. Reduce scope if needed 4. CEO to explore contractor support

Residual Risk: Low (after mitigation)

3.5 B06: Scope Creep

Risk: Continuous feature additions delay delivery.

Attribute	Value
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Probability	**High (60-70%)**
Impact	**Medium**
Risk Score	**HIGH**
Risk Owner	**CTO**

Scenario: - "Just one more feature" syndrome - Feedback leads to constant pivots - Perfectionism delays launch - Phase 2 features creep into Phase 1

Indicators: - Feature list grows after Phase 0 - Milestones consistently missed - "Almost done" state persists - Technical debt accumulates

Mitigation Strategy:

Strategy	Action	Effort
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Feature Freeze	**Lock features after Phase 0 validation**	**Policy**
MVP Mindset	**Ship minimal viable, iterate later**	**Discipline**
Parking Lot	**Document "nice to haves" for Phase 2**	**Process**
Weekly Scope Review	**CEO/CTO review scope weekly**	**30 min/week**

Scope Control Process:

1. All features documented in PRD 2. New requests go to "Phase 2 Parking Lot" 3. Only security/blocking bugs added post-freeze 4. CEO approval required for any scope change

Residual Risk: Low (after mitigation)

3.6 T05: Chrome Extension Rejection

Risk: Chrome Web Store rejects the extension.

Attribute	**Value**
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Probability	**Low (20-30%)**
Impact	**High**
Risk Score	**Medium**
Risk Owner	**CTO**

Scenario: - Extension violates Chrome policies - "Manipulating page content" flagged - Permission requests deemed excessive - Review process delayed

Indicators: - Rejection email from Chrome - Policy violation cited - Required changes significant

Mitigation Strategy:

Strategy	Action	Effort
Policy Review	**Study Chrome extension policies early**	**1 day**
Minimal Permissions	**Request only essential permissions**	**Design**
Transparent Description	**Clear store listing explaining functionality**	**2 hours**
Firefox Backup	**Simultaneously submit to Firefox Add-ons**	**+1 week**

Contingency Plan: If Chrome rejects: 1. Appeal with detailed explanation 2. Modify extension to comply 3. Launch on Firefox first 4. Consider side-loading / enterprise distribution

Residual Risk: Low (after mitigation)

3.7 O01: Cursor Pro Limits

Risk: Cursor Pro's 500 fast requests/month limit slows development.

Attribute	Value
Probability	**Medium (50-60%)**

Impact	**Medium**

Risk Score	**Medium**

Risk Owner	**CTO**

Scenario: - Intensive coding sessions deplete quota mid-month - Slow requests are too slow for iterative development - Productivity drops 20-30% during heavy coding

Indicators: - Hitting quota warnings - Resorting to slow requests frequently - Frustration with tool limitations

Mitigation Strategy:

Strategy	Action	Effort
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Batch Work	**Plan intensive work for start of month**	**Discipline**
Slow Request Usage	**Use slow requests for routine tasks**	**Habit**
Manual Coding	**Fall back to manual when quota depleted**	**Skill**
Upgrade Trigger	**If productivity <70%, request Ultra budget**	**Decision**

Upgrade Decision Matrix:

Situation	Action
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Hitting limit occasionally	**Continue with workarounds**

Hitting limit every month	**Evaluate Ultra upgrade**

Productivity <70%	**Request Ultra budget from CEO**

Residual Risk: Low (after mitigation)

3.8 O05: Security Breach Before Audit

Risk: Security vulnerability exploited before formal audit.

Attribute	**Value**
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Probability	**Low (15-25%)**
Impact	**Critical**
Risk Score	**Medium**
Risk Owner	**CTO**

Scenario: - Ed25519 implementation has subtle flaw - Key management vulnerability - API injection attack - Merchant/user data exposed

Indicators: - Unusual API activity - Reports of invalid signatures - Data access from unknown sources

Mitigation Strategy:

Strategy	**Action**	**Effort**

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Library Usage	**Use well-audited crypto libraries (ed25519-dalek)**	**Design**

Self Review	**Thorough code review of all security code**	**Ongoing**

Open Source	**Community review once open-sourced**	**Month 3**

Minimal Data	**Don't store sensitive user data**	**Design**

Bug Bounty	**Consider informal bug bounty post-launch**	**Launch**

Security Checklist:

- [ ] Use audited crypto libraries only - [ ] No custom crypto implementations - [ ] Input validation on all endpoints - [ ] No secrets in code/logs - [ ] HTTPS everywhere - [ ] Rate limiting on APIs - [ ] Key rotation plan documented

Contingency Plan: If breach occurs: 1. Immediate incident response 2. Rotate all keys 3. Notify affected parties 4. Post-mortem and fix 5. Accelerate formal audit

Residual Risk: Low (after mitigation)

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4. Risk Response Summary

4.1 Risks to Avoid

Risk	**Avoidance Strategy**

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Scope creep	**Feature freeze policy**
Budget overrun	**Weekly tracking, hard limits**

4.2 Risks to Mitigate

Risk	**Mitigation Strategy**
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Steganography failure	**Phase 0 validation, hybrid approach**
No merchant adoption	**Developer-first, open source**
Burnout	**Sustainable pace, rest policy**
Security breach	**Library usage, self-review, open source**

4.3 Risks to Monitor

Risk	**Monitoring Approach**
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CLIP API costs	**Weekly spend review**
Free tier limits	**Usage dashboards**
Chrome policies	**Policy change alerts**

Competitor activity	**Market monitoring**

4.4 Risks to Accept

Risk	Rationale
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Rust learning curve	**C/C++ background accelerates learning**
WASM performance	**Can optimize later**
Minor platform incompatibility	**Focus on major platforms**

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5. Risk Monitoring Plan

5.1 Weekly Risk Review

Every Friday: - [ ] Review active risks - [ ] Check indicators - [ ] Update risk scores if needed - [ ] Report high risks to CEO

5.2 Monthly Risk Report

First Monday of month: - [ ] Full risk register review - [ ] Update mitigation progress - [ ] Identify new risks - [ ] Adjust strategies

5.3 Phase Gate Reviews

At each milestone: - [ ] Comprehensive risk assessment - [ ] Go/No-Go decision - [ ] Risk-adjusted timeline - [ ] Budget impact analysis

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6. Escalation Matrix

Risk Level	Response Time	Escalation To
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Critical	**Immediate**	**CEO**
High	**Within 24 hours**	**CEO**
Medium	**Within 1 week**	**Weekly report**
Low	**Monthly review**	**Monthly report**

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7. Document Approval

Role	Name	Signature	Date
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CEO	**Sobuhle Nyambali**		**/_/2026**

CTO	**Matome Mboweni**		**/_/2026**

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